VINOD KUMAR REDDY BANAVASI

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RESEARCH INTERESTS

Drug Discovery · Structural Biology · AI & Multi-omics Integration · Genomic Diagnostics · ML/DL Automation for Molecular Docking · Sequencing Data Analysis

EDUCATION

University of South Florida Aug 2025 – May 2027
Master of Science, Biotechnology | GPA: 3.8 / 4.0 Tampa, FL, USA

- Member, USF Tri Alpha National Honor Society (Academic Excellence)
- Player, USF Intercollegiate Badminton Team

Osmania University
Bachelor of Science, Microbiology, Genetics & Chemistry | GPA: 3.74 / 4.0 Hyderabad, India

PUBLICATIONS

[1] Banavasi V.K.R. *"Gene Editing: The Revolution of CRISPR Technology."* International Journal of Pharmaceutical Research and Applications, Vol. 10, Issue 2, pp. 344–348, Mar–Apr 2025. ISSN: 2456-4494. Impact Factor: 7.429. DOI: 10.35629/4494-1002344348

[2] Banavasi V.K.R. *"Epigenetics, Stem Cells & Personalized Medicine: A New Era in Cancer Therapy."* International Journal of Pharmaceutical Research and Applications, Vol. 10, Issue 2, pp. 349–353, Mar–Apr 2025. ISSN: 2456-4494. Impact Factor: 7.429. DOI: 10.35629/4494-1002349353

RESEARCH EXPERIENCE

Graduate Research Associate — Structural Biology & Drug Discovery Dec 2025 – Present
Yu Chen Lab, University of South Florida Tampa, FL, USA

- **Viral Protease Crystallography:** Screened 10+ crystal conditions for SARS-CoV-2 Mpro mutants (E166V, E166V/T169S, L50F) and EV-A71 2C ATPase/helicase mutants, achieving diffraction-quality crystals; performed seed sampling and prepared samples for synchrotron beamline X-ray data collection.
- **Molecular Docking & Drug Discovery:** Conducted structure-based drug design and virtual screening against BBPSTS, GPSB, and ZBP1 targets; applied matching spheres and dielectric optimization for lead optimization and hit identification.
- Built automated Python/Linux docking pipelines integrating ML/DL models for high-throughput pose filtering, reducing manual review time and enabling systematic lead prioritization across target panels.
- Performed buffer formulation, chemical condition optimization, and cell-based assay development in support of target identification and lead optimization workflows.

Graduate Research Associate — Clinical Sample Research & Gut Biology Dec 2025 – Present
Thao Ho Lab, University of South Florida Tampa, FL, USA

- Processed clinical faecal and serum samples for iron metabolism and microbiome studies; executed quantitative qPCR, RT-PCR, and ELISA assay development protocols for prostaglandin and metabolic marker profiling.
- Conducted iron exposure studies on neonatal intestinal enteroids: organoid culture, immunofluorescence staining, and gene expression analysis to characterize gut biology responses.

- Expanding into NGS-based multi-omics data analysis; applying bioinformatics workflows for sequencing data interpretation and pathway analysis.

Graduate Research Associate — Multi-Laser Flow Cytometry Sep 2025 – Nov 2025

USF Fred Wright Jr. Flow Cytometry Core · Supervisor: Dr. Charles Szekeres, PhD Tampa, FL, USA

- Designed and executed a 6-color multicolor immunophenotyping panel for B-cell maturation (IgD) and immune activation (CD69) analysis in human whole blood — full workflow from antibody panel design to data interpretation.
- Performed fluorescence compensation, gating strategy development, and quantitative data analysis using FlowJo; contributed to immune subset characterization including CD45, CD19, CD3, CD14 markers.

Summer Research Intern Jan 2025 – Jul 2025

Protein Design Pvt. Ltd. Bengaluru, Karnataka, India

- Hands-on protein expression and analysis via Western blotting, gel electrophoresis, ELISA, and Bradford protein quantification assays; supported senior scientists in experimental design and troubleshooting.
- Performed PCR-based gene amplification, recombinant DNA workflows (DNA extraction, ligation, RNA isolation, molecular cloning), and mammalian cell culture maintenance following GLP documentation standards.

RESEARCH PROJECTS

Evaluation of Cytotoxicity, Genotoxicity & Antibacterial Properties of Carbon Dot Nanoformulation

B.Sc. Dissertation · Aurora's Degree & PG College / CSIR-IICT / Saketh Labs, Osmania University Hyderabad, India

- Assessed CHO-K1 cell biocompatibility via MTT assay (IC50: 59.7 µg/mL); performed chromosomal aberration, mitotic index, and micronucleus genotoxicity assays (OECD TG 473 & TG 487).
- Evaluated antibacterial activity against E. coli and S. aureus by resazurin-based 96-well microdilution (MIC: 200 µg/mL); demonstrated broad-spectrum inhibition via bacterial membrane disruption.

Biosimilars: Scientific Basis, Regulatory Landscape & Development Framework

Project · Aurora's Degree & PG College, Osmania University Hyderabad, India

- Comprehensive research on biosimilar development covering structural comparability, PK/PD assessment, immunogenicity evaluation, and global regulatory pathways (FDA, EMA).
- Analyzed manufacturing challenges: recombinant DNA technology, cell culture consistency, and purification QC for biologic drug replication.

TECHNICAL SKILLS

Drug Discovery: Structure-Based Drug Design (SBDD), Target Identification, Hit Identification, Lead Optimization, Fragment-Based Drug Discovery, Cell-Based Assays, High-Throughput Screening Support

Structural Biology: Protein Crystallography, Crystal Seeding & Harvesting, Buffer Formulation, Condition Optimization, Beamline Sample Preparation, X-ray Diffraction

Molecular Biology: qPCR, RT-PCR, RNA/DNA Extraction, Western Blotting, ELISA, Bradford Assay, Molecular Cloning, Protein Purification, Gel Electrophoresis, Gene Expression Analysis

Cell Biology: Mammalian Cell Culture, Organoid/Enteroid Culture, Immunofluorescence Staining, Cell Viability (MTT), RBC Lysis, Aseptic Technique

Flow Cytometry: Multicolor Panel Design (6-color), Immunophenotyping, Gating Strategy, Fluorescence Compensation, FlowJo, Whole Blood Processing

Computational: Molecular Docking, Virtual Screening, Python, Linux, Machine Learning & Deep Learning, Pipeline Automation, GNINA, ProLIF, RDKit, MDAnalysis, Bioinformatics

Clinical Research: Clinical Sample Handling (Faeces/Serum), Iron Metabolism, Prostaglandin Analysis, Microbiome Research, Multi-omics

Professional: Independent Researcher, Cross-Functional Collaboration, Scientific Communication, Critical Thinking, Problem Solving, Attention to Detail, Adaptability, Time Management, Leadership

FUTURE ASPIRATIONS

Entrepreneurship & Genomic Diagnostics: Long-term goal to build an affordable AI-driven genomic diagnostic platform — using sequencing data, mutation rate patterns, and ML to detect disease risk before symptom onset, and translate insights into personalized health and lifestyle guidance.

PhD: Actively pursuing doctoral opportunities in structural biology, computational drug discovery, or genomics to build a rigorous scientific foundation for translational innovation.

VOLUNTEER & LEADERSHIP

NSS (National Service Scheme) Coordinator

Osmania University Hyderabad, India

- Medical camps for elderly communities; disease awareness programs; COVID-19 food relief; village sanitation drives; forest conservation service; temple food service programs.

Volunteer — Bulls of Moffitt

Moffitt Cancer Center · University of South Florida Tampa, FL, USA

- Crafted and delivered handwritten encouragement cards to cancer patients to provide positivity and emotional support during treatment.

EXTRACURRICULAR & INTERESTS

- Competitive badminton — USF intercollegiate team; district-level competitive swimmer.
- Practitioner of yoga, meditation, and Vedic philosophy; personal interest in bridging ancient pattern-based systems with modern predictive biology and diagnostics.